program 15

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15.Create multiple histogram using ggplot2 facet_wrap() , a density curve representing the probability density,to generate basic box plot using notch and outliers , violin plot,many dotplots,co-realtion matrix to visualize how a variable is distributed to different groups in a built-in R dataset.

Step 1: Load the library

```
library(ggplot2)
```

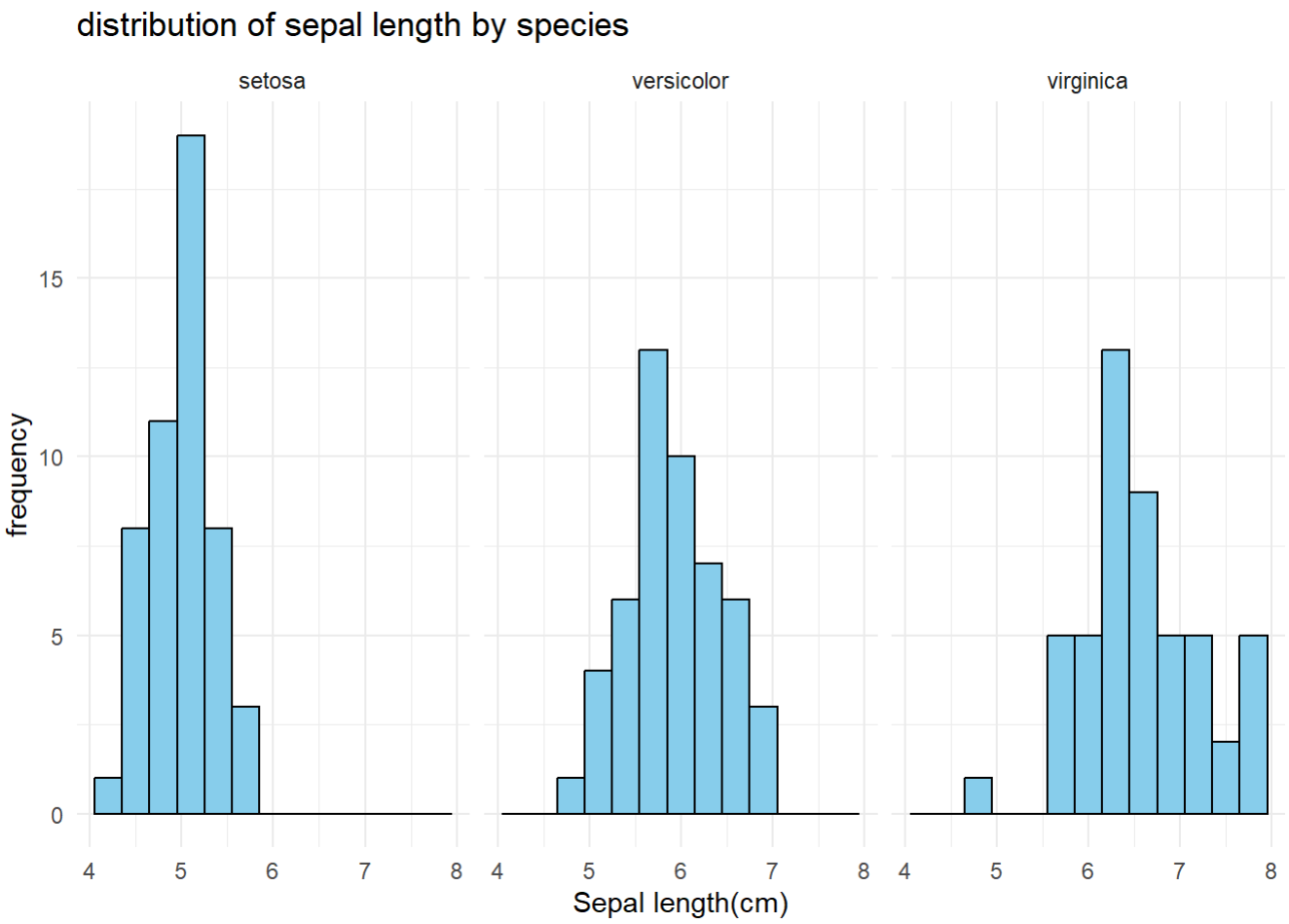
Step 2: Load the built-in dataset

```
data(iris)
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

Step 3: Creating histogram layer

```
ggplot(iris , aes(x = Sepal.Length))+ geom_histogram(binwidth = 0.3,fill = "skyblue",color = "
```



step 4:Define the function

```
plot_density_by_group <- function(data,continous_var , group_var , fill_colors = NULL){
  if(!(continous_var %in% names(data)) || !(group_var %in% names(data))){    stop("invalid col
  }
  p <- ggplot(data, aes_string(x = continous_var, color = group_var, fill = group_var))+
    geom_density(aplha = 0.4)+
    labs( title = paste("density plot of", continous_var,"by",group_var),  x = "continous_var"
    theme_minimal()
  if(!is.null(fill_colors)) {
    p <- p+scale_fill_manual(values = fill_colors)+
      scale_color_manual(values = fill_colors)
  }
  return(p)
}
```

step 5: Load the dataset

```
data("iris")
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
```

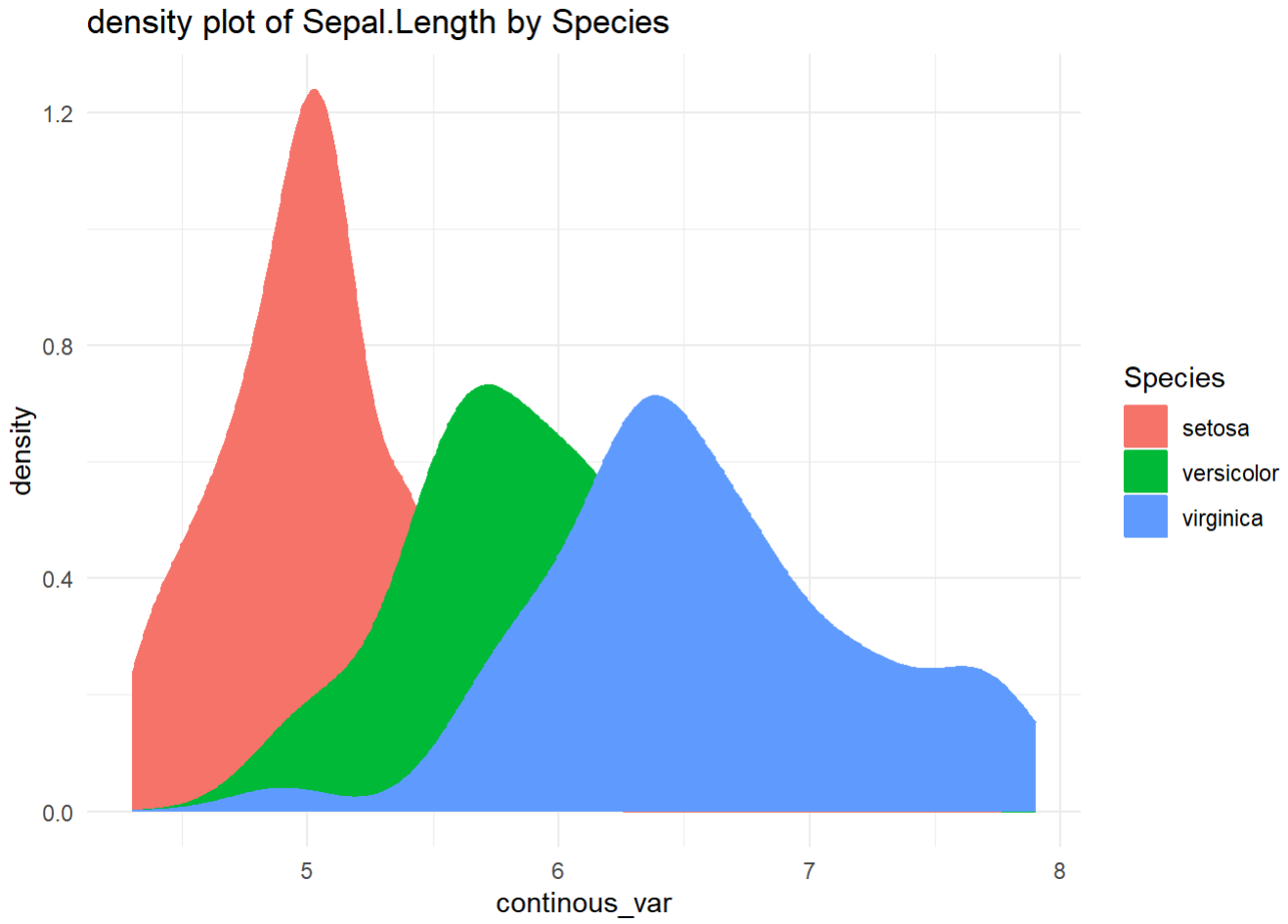
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

Step 6: call the function

```
plot_density_by_group(iris,"Sepal.Length","Species")
```

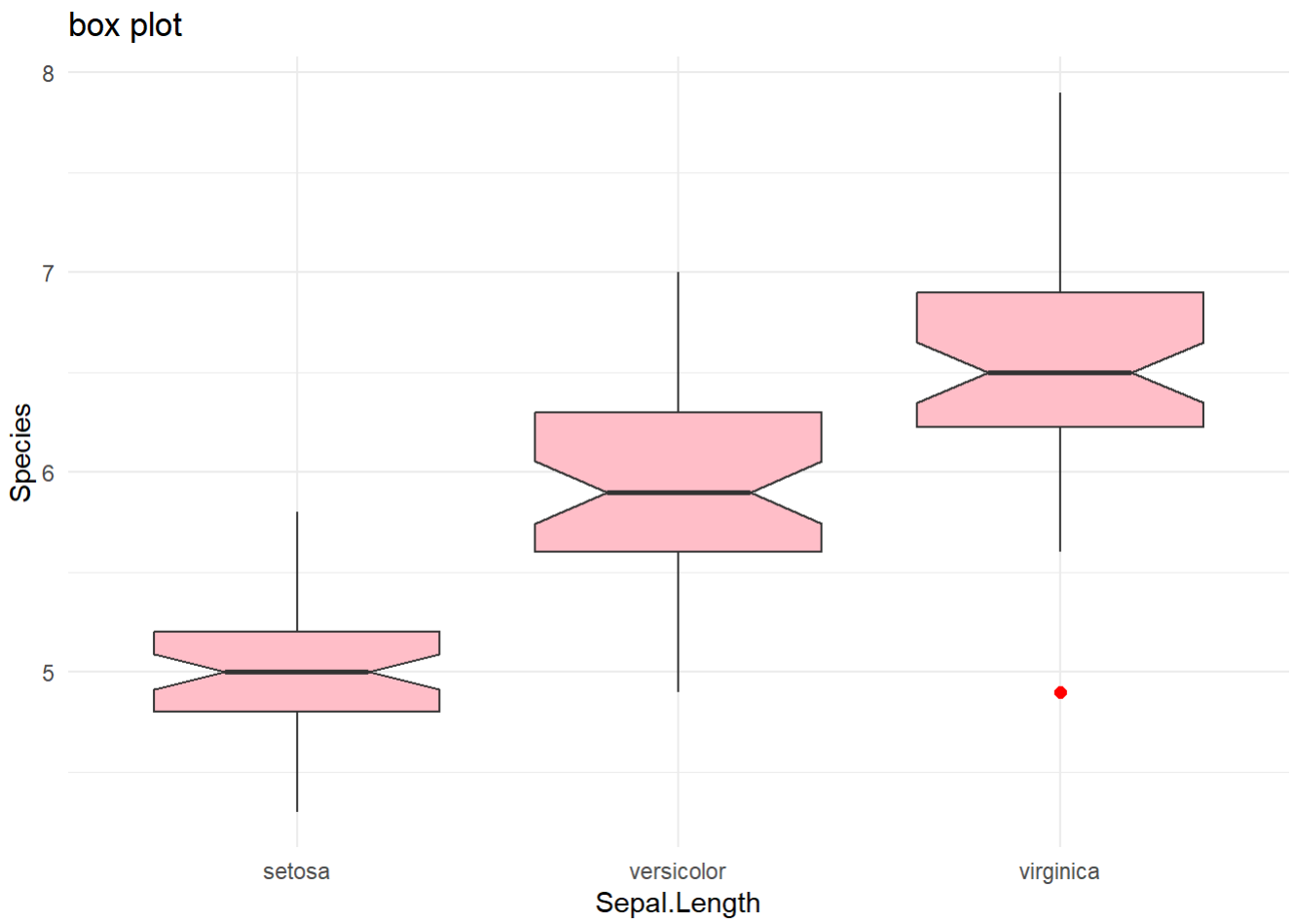
```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
ℹ Please use tidy evaluation idioms with `aes()`.
ℹ See also `vignette("ggplot2-in-packages")` for more information.

Warning in geom_density(aplha = 0.4): Ignoring unknown parameters: `aplha`
```



Step 7:adding graph/layer

```
ggplot(iris,aes(x = Species, y = Sepal.Length))+
  geom_boxplot(notch = TRUE, outlier.color = "red", outlier.shape = 16, outlier.size = 2, fill
  labs(title = "box plot",
       x = "Sepal.Length",
       y = "Species")+
  theme_minimal()
```

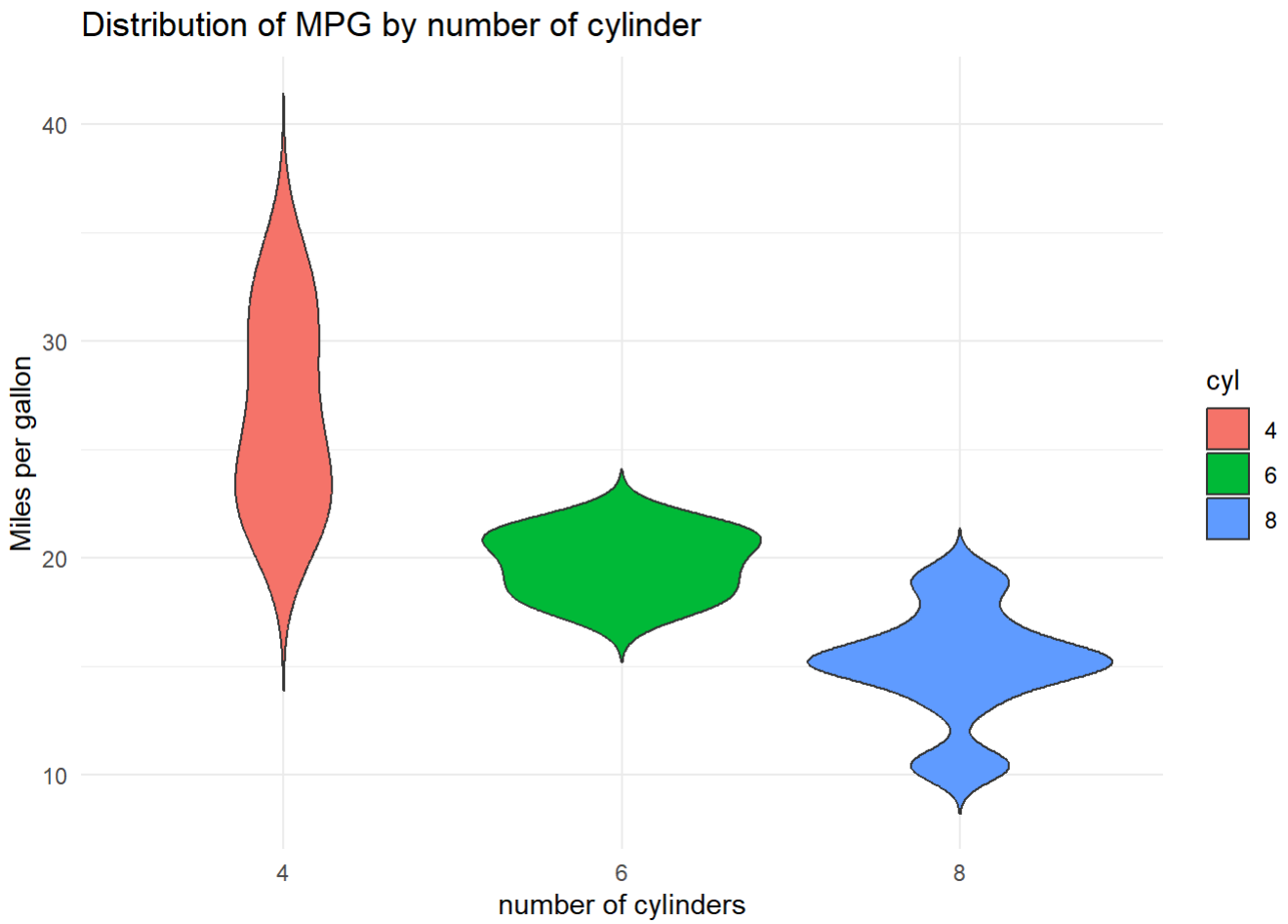


Step 8 : grouping variable

```
mtcars$cyl <- as.factor(mtcars$cyl)
```

Step 9: Create a violin plot

```
ggplot(mtcars, aes(x = cyl,y=mpg,fill = cyl))+
  geom_violin(trim = FALSE)+
  labs( title = "Distribution of MPG by number of cylinder", x = "number of cylinders", y = "M
  theme_minimal()
```



Step 10: load library

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

Step 11: load dataset

```
head("ToothGrowth")
```

```
[1] "ToothGrowth"
```

```
table(ToothGrowth$dose)
```

```
0.5   1   2
```

```
 20  20  20
```

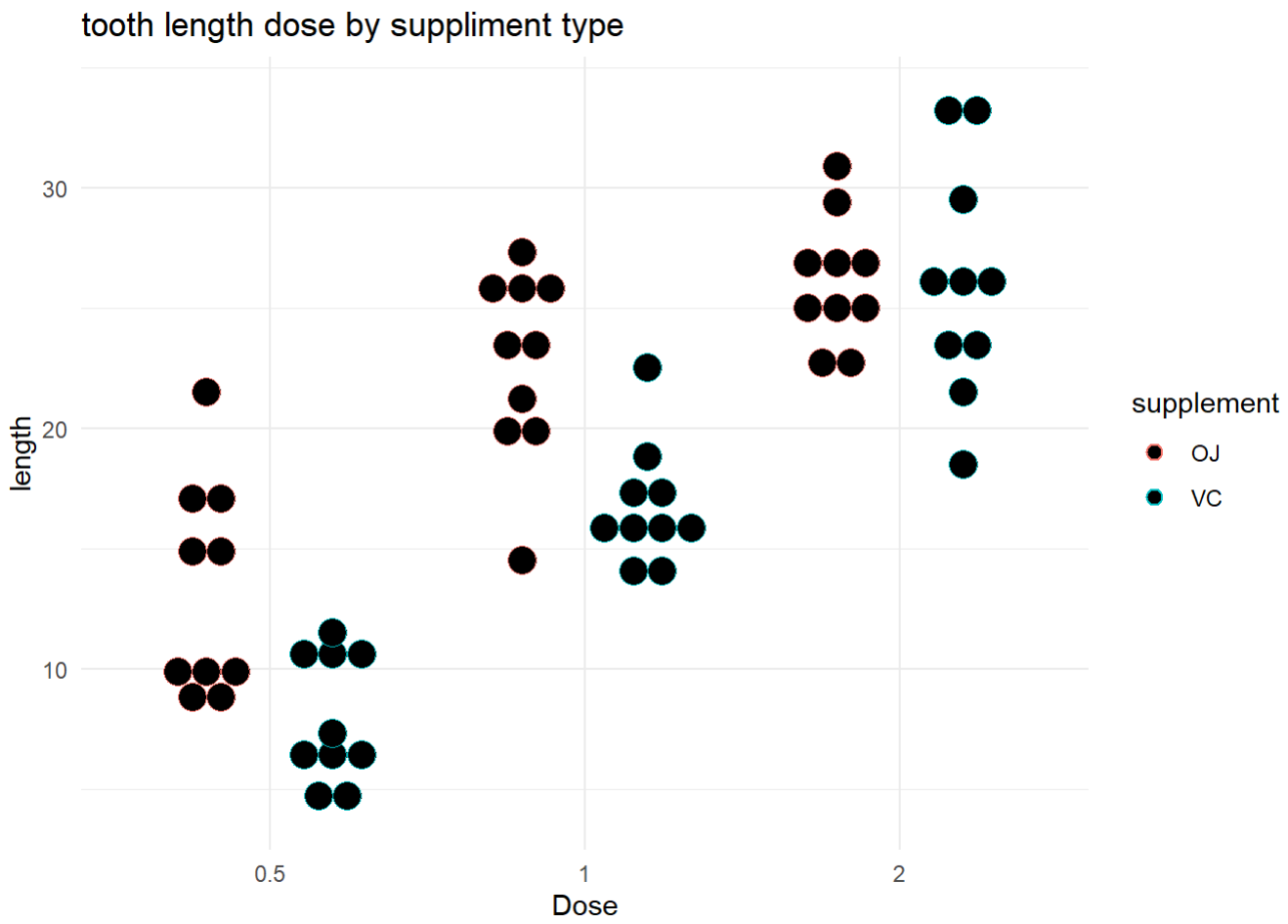
```
class(ToothGrowth$dose)
```

```
[1] "numeric"
```

```
summary(ToothGrowth)
```

```
      len        supp         dose
 Min.   : 4.20   OJ:30   Min.   :0.500
 1st Qu.:13.07   VC:30   1st Qu.:0.500
 Median :19.25           Median :1.000
 Mean   :18.81           Mean   :1.167
 3rd Qu.:25.27           3rd Qu.:2.000
 Max.   :33.90           Max.   :2.000
```

```
ToothGrowth$dose <- as.factor(ToothGrowth$dose)
ggplot(ToothGrowth,aes(x = dose,y = len,color=supp))+
  geom_dotplot(binaxis = "y",
               stackdir = "center",
               position = position_dodge(width = 0.8),
               dotsize = 0.8,
               binwidth = 1.5)+
  labs(
    title = "tooth length dose by suppliment type",
    x = "Dose",
    y = "length",
    color = "supplement"
  )+
  theme_minimal()
```



Step 12: Correlation matrix

```
library(ggplot2)
library(tidyr)
library(dplyr)
head(mtcars)
```

```
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

```
data(mtcars)
cor_matrix <- cor(mtcars)
cor_matrix
```

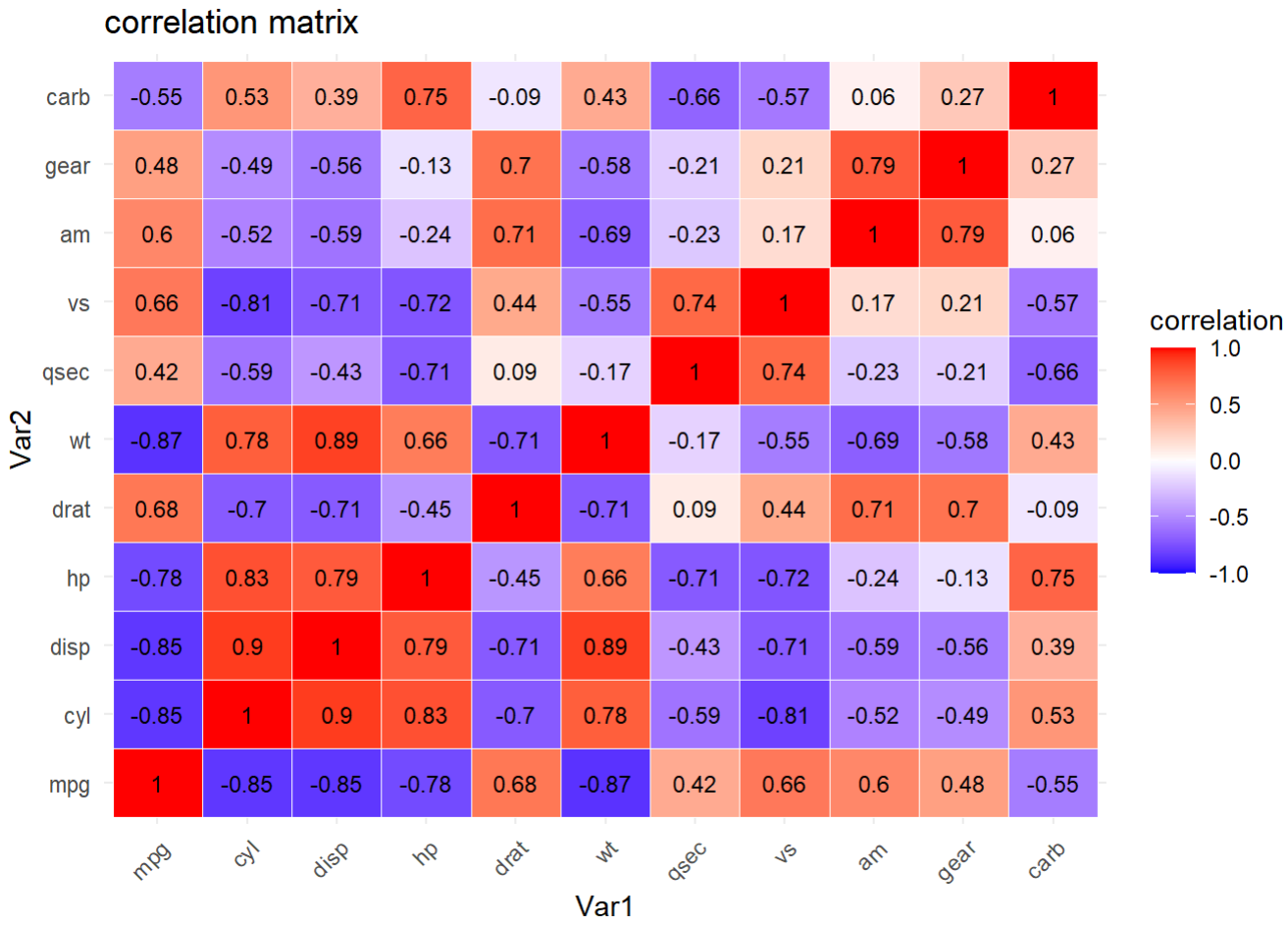
```
            mpg        cyl       disp         hp        drat         wt
mpg   1.0000000 -0.8521620 -0.8475514 -0.7761684  0.68117191 -0.8676594
cyl  -0.8521620  1.0000000  0.9020329  0.8324475 -0.69993811  0.7824958
disp -0.8475514  0.9020329  1.0000000  0.7909486 -0.71021393  0.8879799
hp   -0.7761684  0.8324475  0.7909486  1.0000000 -0.44875912  0.6587479
drat  0.6811719 -0.6999381 -0.7102139 -0.4487591  1.00000000 -0.7124406
```

```
wt   -0.8676594  0.7824958  0.8879799  0.6587479 -0.71244065  1.0000000
qsec  0.4186840 -0.5912421 -0.4336979 -0.7082234  0.09120476 -0.1747159
vs    0.6640389 -0.8108118 -0.7104159 -0.7230967  0.44027846 -0.5549157
am    0.5998324 -0.5226070 -0.5912270 -0.2432043  0.71271113 -0.6924953
gear  0.4802848 -0.4926866 -0.5555692 -0.1257043  0.69961013 -0.5832870
carb -0.5509251  0.5269883  0.3949769  0.7498125 -0.09078980  0.4276059
            qsec         vs          am       gear        carb
mpg   0.41868403  0.6640389  0.59983243  0.4802848 -0.55092507
cyl  -0.59124207 -0.8108118 -0.52260705 -0.4926866  0.52698829
disp -0.43369788 -0.7104159 -0.59122704 -0.5555692  0.39497686
hp   -0.70822339 -0.7230967 -0.24320426 -0.1257043  0.74981247
drat  0.09120476  0.4402785  0.71271113  0.6996101 -0.09078980
wt   -0.17471588 -0.5549157 -0.69249526 -0.5832870  0.42760594
qsec  1.00000000  0.7445354 -0.22986086 -0.2126822 -0.65624923
vs    0.74453544  1.0000000  0.16834512  0.2060233 -0.56960714
am   -0.22986086  0.1683451  1.00000000  0.7940588  0.05753435
gear -0.21268223  0.2060233  0.79405876  1.0000000  0.27407284
carb -0.65624923 -0.5696071  0.05753435  0.2740728  1.00000000
```

```r
cor_df <- as.data.frame(as.table(cor_matrix))
head(cor_df)
```

```
  Var1 Var2       Freq
1  mpg  mpg  1.0000000
2  cyl  mpg -0.8521620
3 disp  mpg -0.8475514
4   hp  mpg -0.7761684
5 drat  mpg  0.6811719
6   wt  mpg -0.8676594
```

```r
ggplot(cor_df,aes( x = Var1 , y = Var2, fill = Freq))+
  geom_tile(color = "white")+
  scale_fill_gradient2(
    low = "blue", mid = "white",high = "red",
    midpoint = 0, limit = c(-1,1),
    name ="correlation"
  )+
  geom_text(aes(label = round(Freq,2)), size = 3)+
  theme_minimal()+
  labs(title = "correlation matrix",
       x = "Var1",
       y = "Var2")+
  theme(axis.text.x = element_text(angle = 45,hjust = 1))
```


correlation matrix
carb -0.55 0.53 0.39 0.75 -0.09 0.43 -0.66 -0.57 0.06 0.27 1
gear 0.48 -0.49 -0.56 -0.13 0.7 -0.58 -0.21 0.21 0.79 1 0.27
am 0.6 -0.52 -0.59 -0.24 0.71 -0.69 -0.23 0.17 1 0.79 0.06
vs 0.66 -0.81 -0.71 -0.72 0.44 -0.55 0.74 1 0.17 0.21 -0.57
qsec 0.42 -0.59 -0.43 -0.71 0.09 -0.17 1 0.74 -0.23 -0.21 -0.66
wt -0.87 0.78 0.89 0.66 -0.71 1 -0.17 -0.55 -0.69 -0.58 0.43
drat 0.68 -0.7 -0.71 -0.45 1 -0.71 0.09 0.44 0.71 0.7 -0.09
hp -0.78 0.83 0.79 1 -0.45 0.66 -0.71 -0.72 -0.24 -0.13 0.75
disp -0.85 0.9 1 0.79 -0.71 0.89 -0.43 -0.71 -0.59 -0.56 0.39
cyl -0.85 1 0.9 0.83 -0.7 0.78 -0.59 -0.81 -0.52 -0.49 0.53
mpg 1 -0.85 -0.85 -0.78 0.68 -0.87 0.42 0.66 0.6 0.48 -0.55
Var2
mpg cyl disp hp drat wt qsec vs am gear carb
Var1
correlation
1.0
0.5
0.0
-0.5
-1.0